

ETIOLOGY AND PATHOGENESIS

Advances are taking place in understanding the pathogenesis of acanthosis nigricans. Insulin clearly plays a central role in the presentation of acanthosis nigricans. In a subset of women with hyperandrogenism and insulin resistance with acanthosis nigricans, loss of function mutations in the insulin receptor or anti-insulin receptor antibodies can be found (Type A and Type B insulin resistance syndromes). It is postulated that excess growth factor stimulation in the skin causes the aberrant proliferation of keratinocytes and fibroblasts, resulting in the characteristic phenotype of acanthosis nigricans. In states of insulin resistance and hyperinsulinemia, acanthosis nigricans may result from excess insulin binding to IGF-1 receptors on keratinocytes and fibroblasts. IGF-1 receptors are expressed on basal keratinocytes and are upregulated in proliferative skin conditions.

Studies show that high concentrations of insulin stimulate fibroblast proliferation through IGF-1 receptors in vitro. Other members of the tyrosine kinase receptor family, including the epidermal growth factor receptor (EGFR) and fibroblast growth factor receptor (FGFR), have also been implicated in acanthosis nigricans. Several genetic syndromes—such as Crouzon syndrome and SADDAN (severe achondroplasia with developmental delay and acanthosis nigricans)—are caused by mutations in fibroblast growth factor receptor 3 (FGFR3) and result in acanthosis nigricans in the absence of hyperinsulinemia or obesity, implicating this growth factor receptor in the disease's pathogenesis. In cases of malignant acanthosis nigricans, transforming growth factor- α (TGF- α) released by tumor cells may stimulate keratinocyte proliferation via activation of the epidermal growth factor receptor.

Support for the role of various growth factors in the pathogenesis of acanthosis nigricans continues to grow.

In addition to direct effects of hyperinsulinemia on keratinocytes, insulin also appears to augment androgen levels in women. High insulin levels stimulate ovarian androgen production and can lead to ovarian hypertrophy with cystic changes. Although acanthosis nigricans is commonly associated with elevated

androgens in women with polycystic ovarian syndrome (PCOS), the skin lesions do not reliably respond to antiandrogen therapy, suggesting that hyperinsulinemia plays a more central role than hyperandrogenism in lesion development. Several drugs have also been reported to cause acanthosis nigricans, including systemic glucocorticoids, nicotinic acid, and estrogens such as diethylstilbestrol.

CLINICAL FINDINGS: CUTANEOUS LESIONS

Clinically, acanthosis nigricans presents as brown to gray-black, papillomatous, hyperpigmented skin thickening in flexural areas, most commonly the posterolateral neck, axillae, groin, and abdominal folds. The distribution is typically symmetric. Affected skin has a velvety, "dirty" appearance.

The back of the neck is the most consistently and severely affected site. Superimposed acrochordons (skin tags) frequently develop in involved areas. In more severe or florid cases, involvement may extend to the dorsal hands, knuckles, and even the palms, where a rugated (ridged) pattern may be observed.

Mucosal surfaces—including oral, esophageal, pharyngeal, laryngeal, conjunctival, and anogenital regions—can also be involved, though this is less common.